

AGL Series Transformer

VAS-AGLST

Compliance with Standards:

- ICAO: Aerodrome Design Manual 9157 Part 5
- FAA AC150/5345-47 (L-830 / L-831, 60 Hz / 50 Hz)
- IS: 12290 - 1987
- IEC 61823 (L-830 / L-831, 60 Hz / 50 Hz)

Features

- VAS offers AGL-ST both earthing type and without earthing type. In the earthing type transformers earthing is connected to the end of the secondary side 4mm socket as shown in Fig - A.
- VAS-300 series transformers usually supplies the current in the Airfield Ground Lighting series circuit that provides electrically separation between the primary and secondary windings and linked well magnetically.
- The transformer is totally encapsulated with Thermoplastic Vulcanizer and are designed in low loss, high magnetic permeability toroidal core to get superior electrical performance.
- The transformer is designed for lowest inductance and magnetic leakage which provides good power factor and higher efficiency and support lamp control and monitoring requirements.
- The transformer is duly insulated and covered with Thermo Plastic Vulcanizer having excellent electrical and mechanical properties and also good chemical resistance too.
- TPV resists weathering, UV radiation and ozone exposure. TPV material also resist the temperature effects upto 130 deg C.
- All contacts such as primary pin and sockets and secondary sockets are duly made up of copper and plated with nickel chromium .
- The sockets are duly supplied with a bronze type sleeve / spring to ensure adequate contact pressure.

Applications

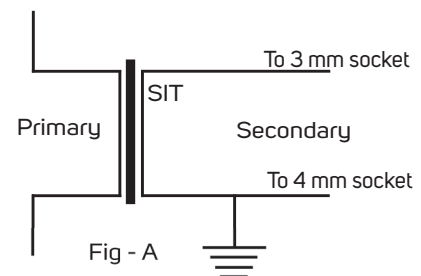
AGLST are used in series airfield lighting circuits to electrically isolate the lighting fixtures from the medium voltage usually present and to ensure circuits continuity in the event of lamp failures too.

Electrical Characteristics of AGL-ST

- Rated power 10 - 300 W
- Rated current 6.6 A/6.6 A
- Rated Voltage 5000V/600V
- Power factor > 0,98
- L (leak) 10μH - 80 μH
- L (magn) 10mH - 50 mH
- Temperature range -25° C to +55° C

Testing

All units (100 percent) are high voltage tested and their ratio confirmed. All ratio testing done with appropriate frequency 50 Hz or 60 Hz for precision. no conversion factor used.



Electrical Information

VAS Type with Earthing	VAS Type without Earthing	Rated power in watts	Power Range in watts	FAA Type	Rated Current in Amps	Load in Ω	Power Factor	Efficiency in %	Inductance Leakage in μ H	Magnetic Leakage in mH	Short Circuited Current	Open Circuited Voltage
VAS 615E	VAS 615	10/15	10-15	L-830-16 L-831-16	6.60 / 6.60	0.34	> 0.98	> 70	< 15	< 10	< 6.70	< 5
VAS 625E	VAS 625	20/25	20-25	L-830-17 L-831-17	6.60 / 6.60	0.57	> 0.98	> 70%	< 15	< 10	< 6.70	< 8
VAS 645E	VAS 645	30/45	25-60	L-830-1 L-831-1	6.60 / 6.60	1.15	> 0.98	> 80%	< 30	< 20	< 6.70	< 12
VAS 665E	VAS 665	65	50-85	L-830-3 L-831-3	6.60 / 6.60	1.60	> 0.98	> 80%	< 30	< 20	< 6.70	< 16
VAS 6100E	VAS 6100	100	80-125	L-830-4 L-831-4	6.60 / 6.60	2.24	> 0.98	> 85%	< 30	< 20	< 6.70	< 25
VAS 6150E	VAS 6150	150	120-180	L-830-18 L-831-18	6.60 / 6.60	3.58	> 0.98	> 90%	< 40	< 25	< 6.70	< 30

Insulation Resistance		
Winding under Test	Voltage Applied (kv DC)	Minimum Insulation Resistance (Megohms)
Hot/Cold Primary for transformers up to 300 W	15.0	> 7500
Hot/Cold Secondary for transformers up to 300 W	5.0	> 2500

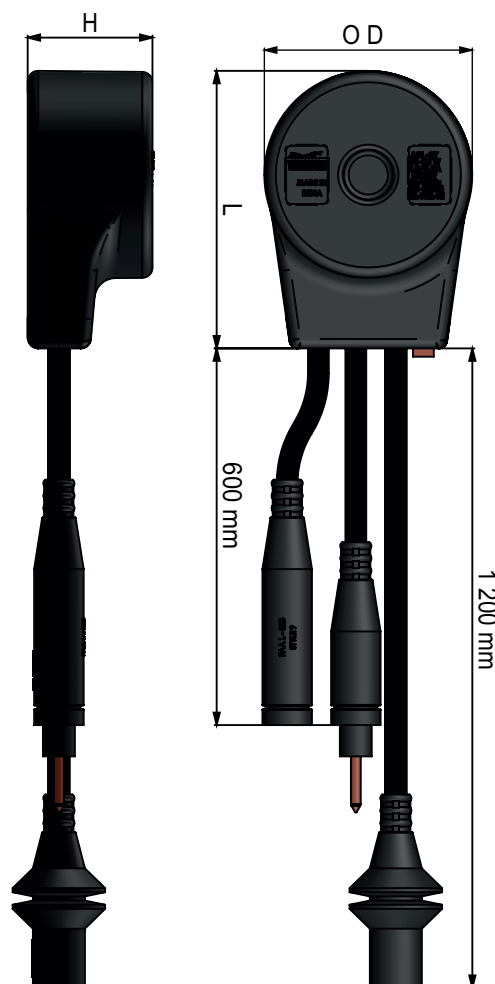
Connections and Materials :

The primary leads are 8.3 sq.mm size with a length of 60cm as per the requirement of FAA L-823, Style 2 Plug and Style 9 receptacle.

The secondary leads are 2.5 sq.mm size with a length of 120cm as per the requirement of FAA Style 8 Receptacle.

Dimensions and Weight

VAS Type with Earthing	VAS Type without Earthing	OD in mm	Length in mm	Height in mm	Weight in kgs
VAS 615E	VAS 615	100	133.5	60	2.2
VAS 625E	VAS 625	100	133.5	60	2.4
VAS 645E	VAS 645	115	148.5	70	2.8
VAS 665E	VAS 665	115	148.5	70	3.0
VAS 6100E	VAS 6100	125	158.5	70	3.5
VAS 6150E	VAS 6150	140	173.5	75	4.2



Insulation Resistance		
Winding under Test	Voltage Applied (kv DC)	Minimum Insulation Resistance (Megohms)
Hot/Cold Primary for transformers up to 300	15.0	> 7500
Hot/Cold Secondary for transformers up to 300	5.0	>2500

Connections and Materials :

The transformer is duly insulated and covered with Thermo Plastic Vulcanizer having excellent electrical and mechanical properties and also good chemical resistance too. TPV resists weathering, UV radiation and ozone exposure. TPV material also resists the temperature effects up to 130 deg C. All contacts such as primary pin and sockets and secondary sockets are duly made up of copper and plated with nickel chromium. The sockets are duly supplied with a bronze type sleeve / spring to ensure adequate contact pressure.

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VAS 6150E	VAS 6150	140	173.5	75	4.2

